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## **Novel Strategies to Improve Production in an Unconventional Jurassic Reservoir with Asphaltene Challenges: Insights from South Kuwait**

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### **Abstract**

The South Kuwait Deep Jurassic formation is classified as an unconventional with abnormal pressure, low matrix permeability, and porosity. This heterogeneous carbonate formation yields light oil but has a significant issue with asphaltene deposition in the wellbore. During the initial field development phase, the field began to suffer from a significant production decline and rapid reservoir pressure depletion. These challenges led to the stimulation of the wells, driven by multiple factors, the desire to connect natural fractures, fully drain the formation, and increase the Estimated Ultimate Recovery (EUR) of the wells by accessing additional stimulated reservoir volume.

Following well candidate selection, developing a solution phase that would achieve the treatment's objectives while addressing reservoir challenges such as asphaltene flocculation, stimulation under depleted conditions, and high fluid leak-off tendency became one of the primary concerns. Upon identification of organic deposits, cost-effective remediation strategies were implemented, including the injection of unique Aromatic-Aliphatic Blends (AAB) to re-dissolve deposited asphaltenes near the wellbore and increase injectivity for stimulation. The wells were then treated using High Viscosity Friction Reducer (HVFR) PAD fluid and a modified single-phase retarded acid system (mSPRAS) to ensure deep and branched ramifications. The chemical diversion system was designed with hydrochloric acid and viscoelastic surfactants to enhance penetration and divert fluid from high-permeability zones into low-permeability zones. Holistic design diagnostics—including memory gauge data, Diagnostic Fracture Injection Test (DFIT) analysis, and swift log analysis for fracture height computation—were conducted to revalidate fracture simulations and fine-tune the design, rather than relying on conventional "cookie-cutter" methods. The treatment design focused on improving fluid loss control while initiating high-conductivity fractures within the reservoir. Strategic measures, including the use of optimized diverters and tailored acid systems, were implemented to maximize fracture length and effectiveness.

Based on comparative evaluation, employing these strategies has resulted in significantly improved performance compared to wells fractured conventionally. Bottom-hole data, fracture analysis, and log data confirmed that the fracture remained within the targeted interval, with an average half-length of 300 ft. In

fact, the productivity index of the wells increased significantly, representing a substantial enhancement in output. Furthermore, the treatment effectively mitigated risks associated with asphaltene deposition.

This paper summarizes the candidate selection criteria, application of this unique stimulation design in Kuwait, operational procedures, diagnostic tools, well clean-up, and the production performance aspects of the treatment. The success of this technique is critical for overcoming production decline and reservoir pressure depletion and for effectively stimulating this prolific unconventional carbonate formation. The insights presented can be applied to similar challenges in other fields.

## Introduction

The South Kuwait Deep Jurassic formation is classified as an unconventional with abnormal pressure, low matrix permeability, and porosity. The Deep reservoir, situated in Kuwait's southeast region (Figure 1)—represents a tight, heterogeneous carbonate system with complex geomechanical properties and variable petrophysical quality. Porosity generally ranges from moderate to locally high, reflecting pronounced vertical and lateral variations due to interbedded lithologies and diagenetic alterations. Permeability remains low, averaging around 17 millidarcies, confirming its classification as a tight reservoir. The crude oil is light ( $\sim 38^\circ$  API), and production is mainly supported by solution-gas expansion, with limited assistance from the natural aquifer. Over the span, the reservoir pressure has declined reaching approximately less than 10,000 psi.

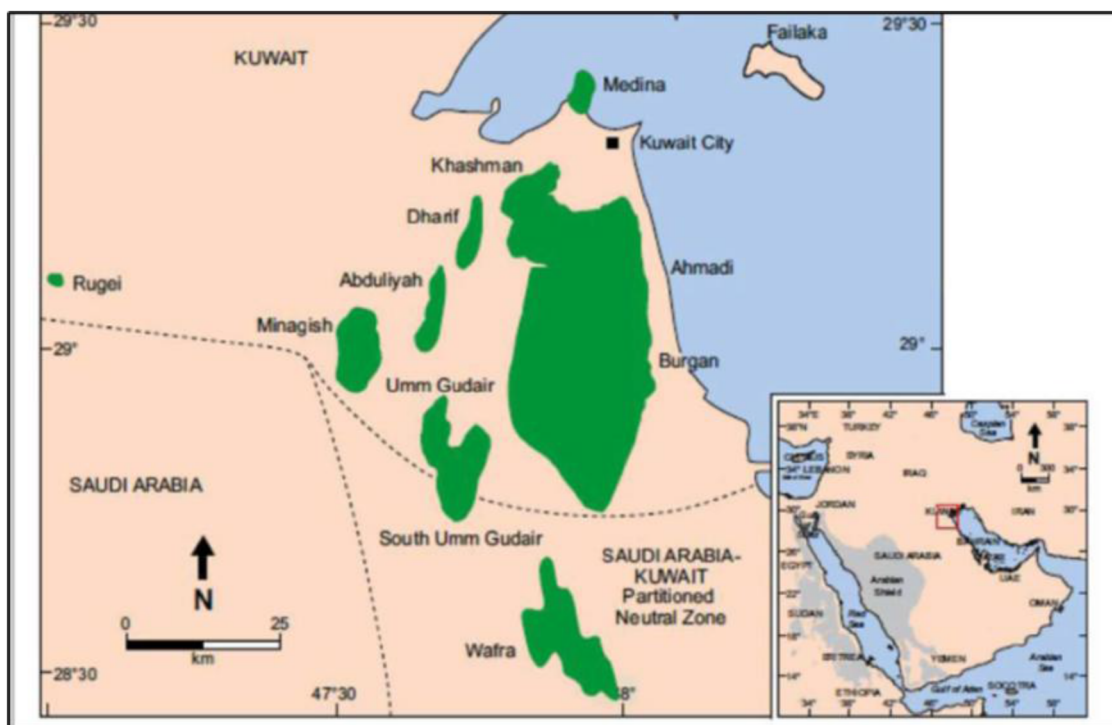


Figure 1—Location of Southeast Kuwait Field

This Jurassic-age high-pressure, high-temperature (HPHT) carbonate oil reservoir presents significant production challenges. While it produces light oil, its productivity is severely impacted by persistent asphaltene deposition within the wellbore and inherently low matrix permeability, both of which hinder efficient hydrocarbon flow and overall reservoir performance.

There is growing concern that asphaltene deposition could pose a risk in the reservoir if the pressure drops below the Asphaltene Onset Pressure (AOP), potentially leading to irreversible damage and total loss of hydrocarbon recovery. At present, the reservoir pressure remains near its initial value and is above any

reliably measured AOP in the field. To assess this risk, a flow assurance study was performed to determine the AOP of the reservoir fluid at various temperatures. Since AOP is temperature-dependent and produces fluids cool as they travel from the reservoir to the wellhead, understanding this relationship is critical. Bottom-hole samples from five wells in this field were analyzed, and AOP measurements ranged from 3,900 psia to 8,000 psia (Al-Shammari, A., et al. 2024). Following a stringent quality control process for all reservoir-temperature measurements, it was concluded that the AOP for this reservoir fluid is  $4,300 \pm 500$  psia. The relationship between AOP and temperature is essential for identifying the depth within the tubing where asphaltene may begin to precipitate and accumulate.

Laboratory and field studies on this reservoir indicate a clear temperature dependency of asphaltene onset pressure (AOP): as temperature decreases, AOP increases. Data from multiple wells, supported by EOS modeling, show that the pressure at which asphaltene flocculation begins rises significantly when the temperature drops, with observations extending down to 150 °F. At reservoir conditions (approximately 220–250 °F), AOP values typically range between 5,100 and 6,100 psi, with reported measurements spanning from about 4,400 to 6,500 psi based on gravimetric, SDS/NIR, and HPM/PSA tests (Al-Obaidli, A., 2019). These findings, reinforced by multi-temperature experiments and core-flood studies, have been used to construct AOP–temperature relationships and define the asphaltene flocculation envelope. The practical implication is that as fluids cool while moving up the wellbore, the onset pressure shifts higher, narrowing the safe operating window and increasing the risk of deposition unless mitigated.

Geomechanically, the reservoir is characterized by low natural injectivity and tight-rock behavior, making fracturing essential to improve hydrocarbon recovery. A significant operational consideration is the risk of asphaltene deposition, which can occur if bottomhole pressures fall below the onset level of roughly 6,500 psi. Additionally, the reservoir fluids contain corrosive components—including around 1,000 ppm H<sub>2</sub>S and approximately 4% CO<sub>2</sub>—which must be accounted for in well completion design and surface equipment selection. While vertical connectivity is relatively robust, compartmentalization arises from faults and subtle stratigraphic discontinuities. Among the three units, the Middle unit consistently exhibits the highest productivity, evidenced by sustained production. These reservoir zones exhibit significant variability due to heterogeneity; therefore, geomechanical logging (e.g., sonic logs, density logs) is employed to map these properties across the field. Brittleness is the rock's tendency to fracture rather than deform plastically. In hydraulic fracturing, more brittle rocks are preferred because they fracture more easily and generate complex, productive fracture networks. In this reservoir, zones with lower shale content and higher carbonate composition (limestone or dolomite) generally display higher brittleness and are thus better candidates for stimulation. Young's modulus measures a rock's stiffness—its ability to resist deformation under applied stress—and is defined as the ratio of stress to strain within the elastic regime. Higher Young's modulus indicates stiffer rocks capable of storing more elastic energy, producing wider fractures during hydraulic stimulation. The Young's modulus typically ranges between 2.5 to 4 million psi depending on lithology and porosity, with higher values associated with competent limestone or dolomite intervals. Poisson's ratio, which describes the ratio of lateral strain to axial strain under compression, is another key indicator of brittleness. Lower Poisson's ratio values (e.g., 0.15–0.25) are usually linked with fracture-prone rocks. In this zone (Figure 2) combining lower Poisson's ratio with higher Young's modulus represents the most favorable targets for hydraulic fracturing operations.

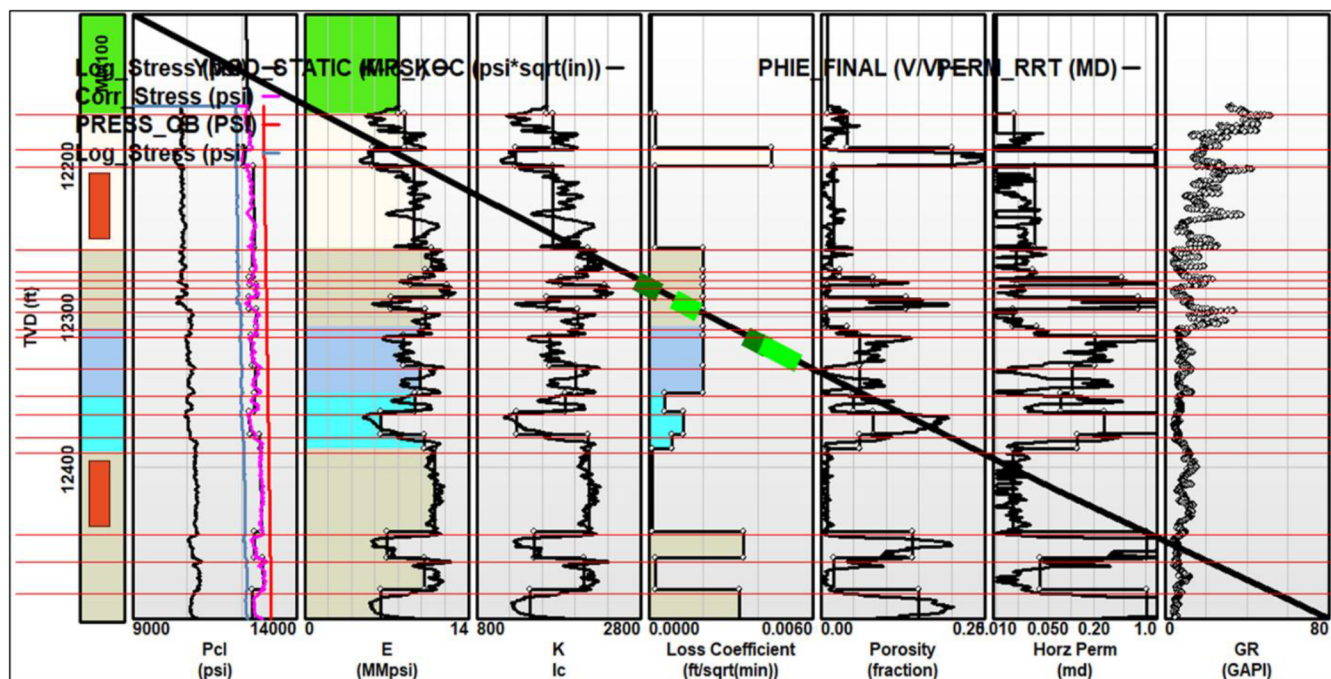


Figure 2—Well A Geomechanical Properties

## Completion Strategy

A critical factor in achieving successful treatment is the careful identification of appropriate candidate wells. Assessing a well's suitability requires consideration of multiple parameters, and without a clear understanding of the treatment objectives, it is difficult to pinpoint wells with the greatest potential for economic benefit. The main goal is to improve the Estimated Ultimate Recovery (EUR) by expanding the volume of the reservoir that is effectively stimulated.

Well-A, situated in this field, targets a formation with substantial potential. The reservoir pressure in this zone is around 0.9 psi/ft. The well was drilled and completed with a 5½-inch casing, and specific intervals were chosen, as shown in Figure 3. In this section, the formation has an approximate vertical thickness of 120 feet. Key reservoir and well parameters are summarized in Table 1. Following a detailed assessment, Well-A was deemed a suitable candidate for acid hydraulic fracturing.

Table 1—Well A Properties

| Parameters                                               | Value              | UOM     |
|----------------------------------------------------------|--------------------|---------|
| Fluid Type                                               | Light Oil (36 deg) | oAPI    |
| Deviation Averaged                                       | 38°                | o       |
| Initial GOR                                              | 900                | scf/bbl |
| H <sub>2</sub> S                                         | ~200               | ppm     |
| CO <sub>2</sub>                                          | 0.52               | wt%     |
| Reservoir Pressure                                       | 12009              | psi     |
| Reservoir Temperature                                    | 229                | oF      |
| *Formation Permeability<br>(*Natural Fractures + Matrix) | 2.415              | mD      |
| Initial GOR                                              | 900                | scf/bbl |
| Fracturing Pressures                                     | 12718              | psi     |

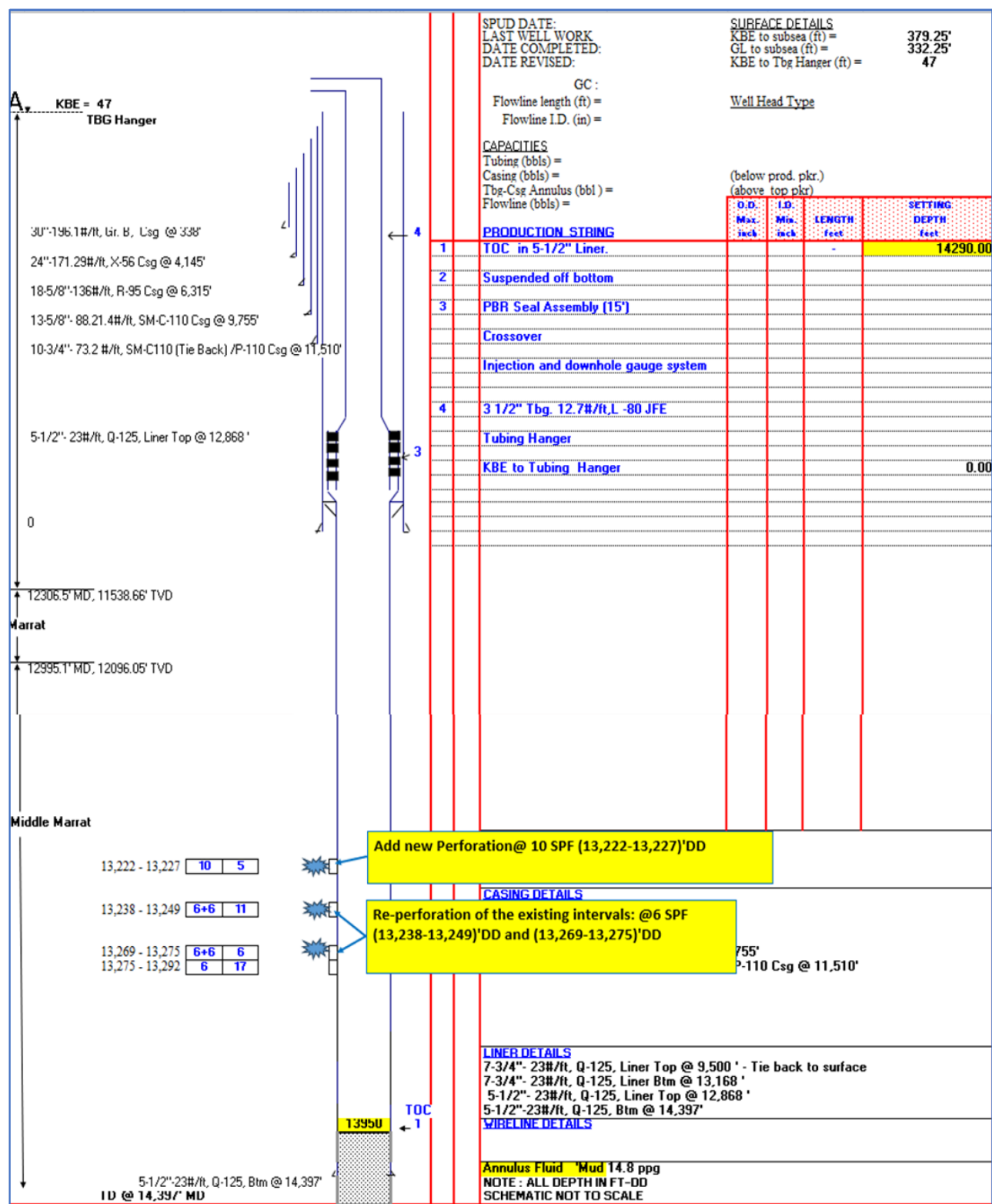


Figure 3—Well A Schematic

## Well A Strategy

### DFIT (Diagnostic Fracture Injection Testing)

Pre-DFIT Injection of Aromatic-Aliphatic Blends (AAB), as in Table 2 is used to restore near-wellbore injectivity in this reservoir as it is affected by asphaltene deposition. Asphaltenes, high molecular weight components in crude oil, tend to precipitate under pressure drop or fluid incompatibility, blocking pore spaces and reducing permeability. The unique aromatic fraction of AAB dissolves these asphaltene deposits by disrupting intermolecular forces, while the aliphatic fraction lowers oil viscosity and enhances fluid mobility. This combined action reopens pore channels, improves permeability, and allows stimulation fluid to penetrate deeper into the formation, thereby increasing the effectiveness and overall injectivity.

Table 2—DFIT Schedule

| Step Number | Step Name                                                                        | Slurry Rate (bpm) | Fluid Name | Slurry Volume (bbl) | Total Slurry (bbl) | Pump Time (min) | Cum Time (min) |
|-------------|----------------------------------------------------------------------------------|-------------------|------------|---------------------|--------------------|-----------------|----------------|
| 1           | Open W/H and Record Surface Pressure.                                            |                   |            |                     |                    |                 |                |
| 2           | AAB                                                                              | 3                 | AAB        | 30                  | 30                 | 10              | 10             |
| 3           | Displacement                                                                     | 3                 | 2% KCL     | 150                 | 180                | 50              | 60             |
| 4           | Stop Pumping & Record Pressure till Pressure Stabilization to Open W/H Pressure. |                   |            |                     |                    |                 |                |
| 5           | Spearhead                                                                        | 3                 | 15% HCL    | 50                  | 230                | 17              | 77             |
| 6           | Displacement                                                                     | 3                 | 2% KCL     | 120                 | 350                | 40              | 117            |
| 7           | Stop Pumping & Record Pressure till Pressure Stabilization to Open W/H Pressure. |                   |            |                     |                    |                 |                |
| 8           | Pump till Press $\Delta$                                                         | 5                 | 2% KCL     | 10                  | 360                | 2               | 119            |
| 9           | DFIT-1                                                                           | 5                 | 2% KCL     | 30                  | 390                | 6               | 125            |
| 10          | Record Pressure till Radial flow is achieved. Record for 3 hours.                |                   |            |                     |                    |                 |                |
| 11          | DFIT-2                                                                           | 5                 | 2% KCL     | 30                  | 420                | 6               | 131            |
| 12          | Record Pressure till Closure is achieved. Record for 2 hours.                    |                   |            |                     |                    |                 |                |

The diagnostic fracture injection test was conducted to evaluate fracture leakoff behavior, determine the closure stress gradient, and assess fluid efficiency—key parameters for designing an optimized main hydraulic fracturing (mainfrac) treatment. This test provides critical insights into reservoir properties such as stress, pressure, and permeability, all of which influence fracture efficiency and overall, well performance. The test utilized a base fluid at a relatively low volume but at a high injection rate to maintain dynamic conditions. The fluid volume was carefully selected to ensure adequate fracture coverage across the target pay zone. During the shut-in period, pressure decline was continuously monitored (Figure 4) and analyzed until the system transitioned to pseudo-radial flow, allowing estimation of reservoir transmissibility and storage characteristics. G-function analysis was applied to interpret the pressure falloff data and identify the dominant leakoff mechanism. Results indicated a standard leakoff regime within the formation, with a fracture gradient of 1.00 psi/ft, a closure gradient of 0.97 psi/ft, and a reservoir pressure gradient of 0.9 psi/ft (Figures 5). These findings highlight the importance of precise injection rate control to achieve optimal fracture geometry and maximize half-length propagation during mainfrac operations.

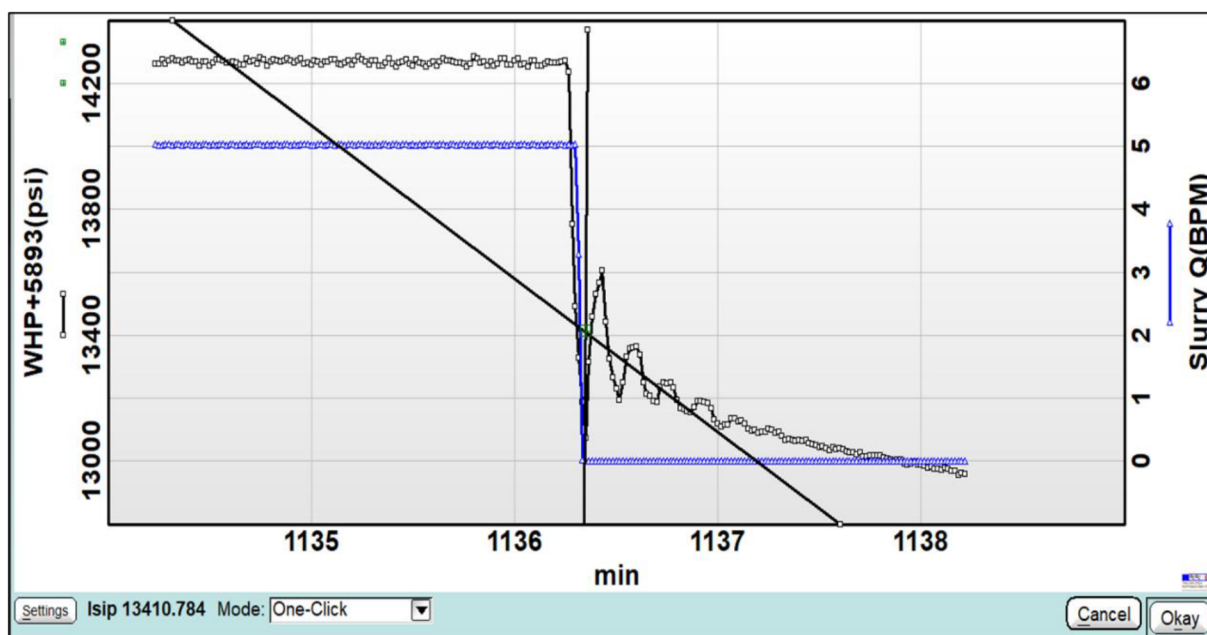


Figure 4—DFIT Analysis- ISIP Plot

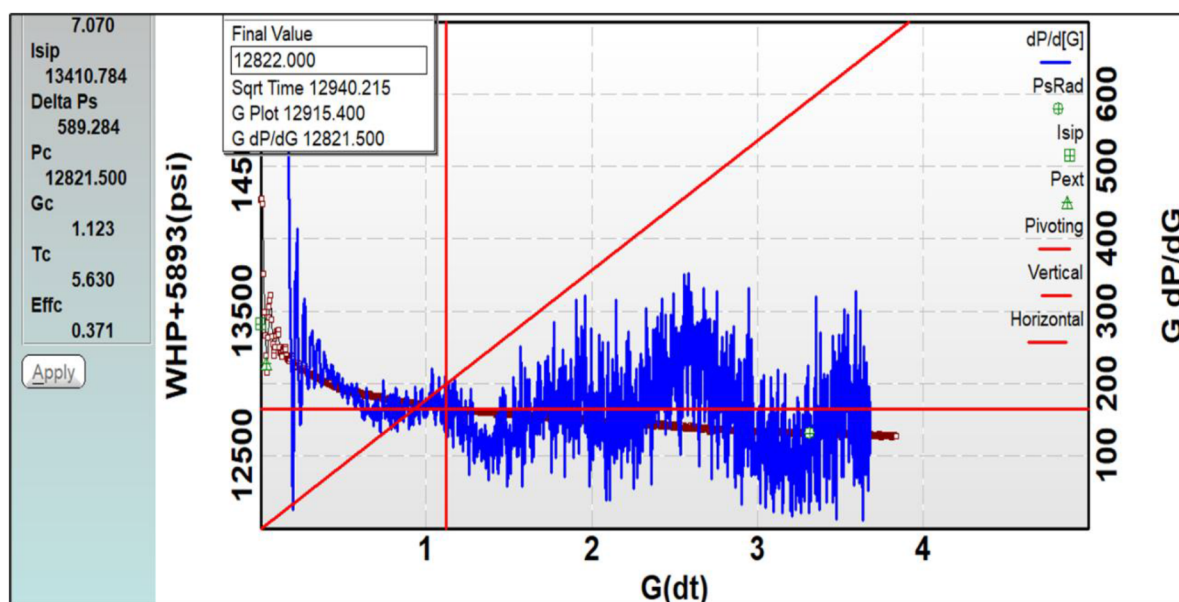


Figure 5—DFIT Analysis- G Function Plot

### Step-rate test (SRT)

The test was conducted to accurately determine the fracture closure pressure and to evaluate both the extension rate and the pressure required for fracture propagation. A step-rate test was implemented, consisting of a series of short, successive pumping stages, each typically lasting one to two minutes or less, with progressively increasing pressure. This approach allows for precise observation of the fracture response to changing injection conditions. In the case of a smooth, planar fracture that opens proportionally to the applied fluid pressure, the net fracture pressure—difference between the bottomhole treating pressure and the closure pressure—shows a direct correlation with the injection rate, as illustrated in Figure 6. This relationship provides insight into the fracture mechanics, enabling the determination of the extension rate and the pressure required to propagate the fracture further into the formation. Analysis of the step-rate data for this well indicated a fracture extension rate exceeding 1.68 bpm and a bottomhole extension pressure of

13,135 psi, as shown in Figure 7. These results highlight the fracture's dynamic response and the hydraulic conditions necessary for effective propagation. By using short-duration pumping stages with incremental pressure increases, the step-rate test reduces uncertainties compared with longer, less controlled injection methods, ensuring more reliable determination of critical fracture parameters.

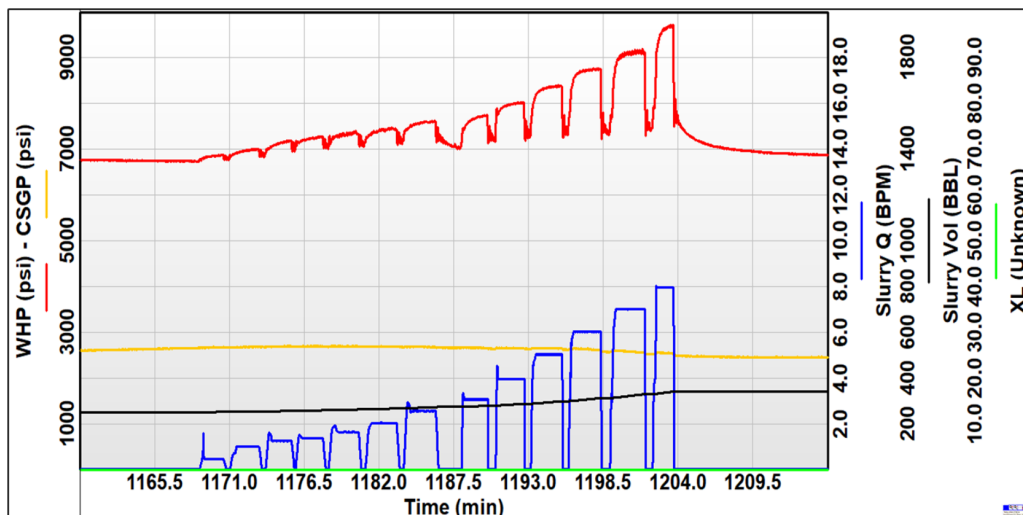


Figure 6—SRT Analysis- Pressure Plot

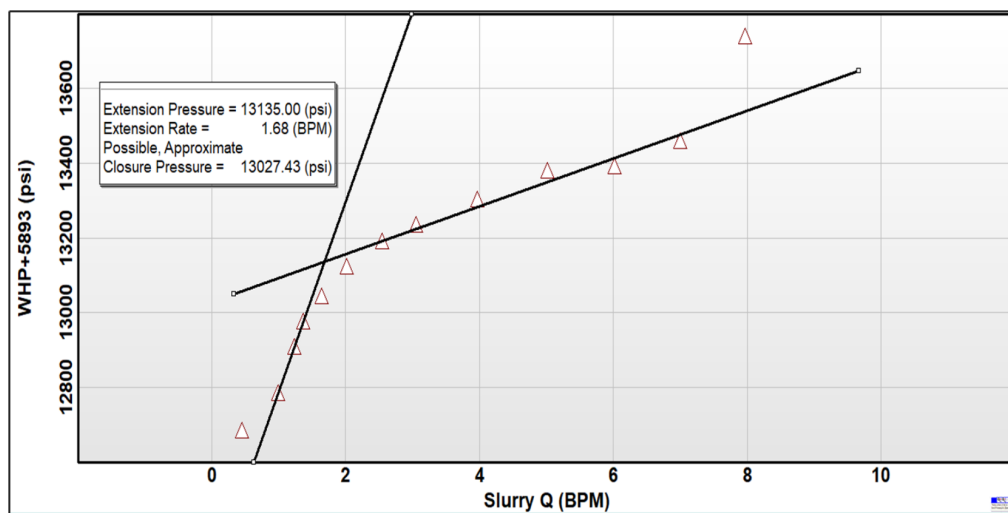


Figure 7—SRT Analysis- Extension Plot

## Minifrac

A minifrac conducted with HVFR fluid (Table 3) is an essential step in hydraulic fracturing design to evaluate fracture propagation, closure pressure, and fluid efficiency. In this test, a small volume of fracturing fluid is injected under controlled conditions, using a pad volume to initiate the fracture. The PAD fluid enhances viscosity, improving acid transport potential in the full-scale treatment, and allows for better fracture containment. Fluid efficiency, defined as the ratio of the volume of fluid entering the fracture to the total injected volume, is usually monitored during the minifrac to assess leak-off and fracture geometry; typical values for HVFR fluids range from 60–80%, indicating effective fracture creation and minimal fluid loss into the formation. Results indicated a standard leakoff regime within the formation, with a fracture gradient of 1.04 psi/ft, a closure gradient of 0.96 psi/ft, and a reservoir pressure gradient of 0.91 psi/ft (Figure 8).

Table 3—Well A Minifrac Schedule

| Step      | Slurry Rate (bpm) | Fluid Name | Slurry Volume (bbl) | Total Slurry Vol. (bbl) | Pump Time (min) |
|-----------|-------------------|------------|---------------------|-------------------------|-----------------|
| Mini Frac | 12                | HVFR       | 150                 | 150                     | 12.5            |
| Flush     | 12                | 2% KCL     | 120                 | 270                     | 10.0            |

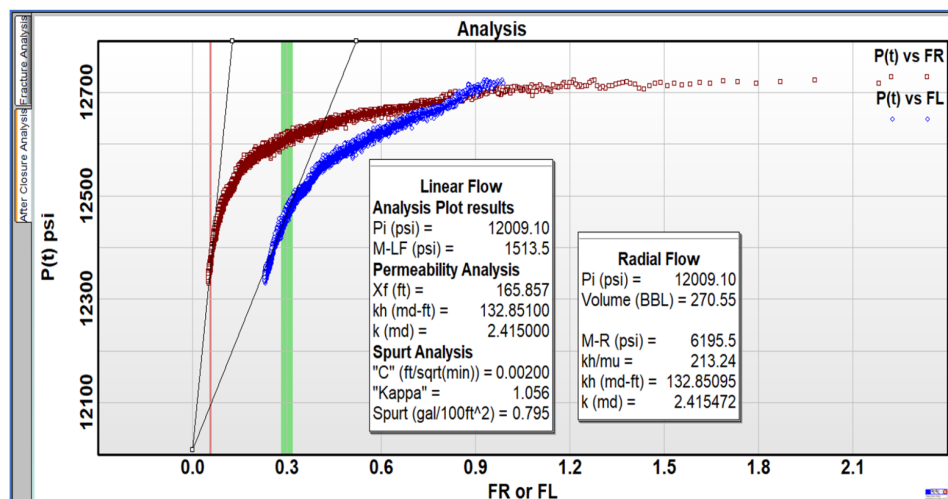
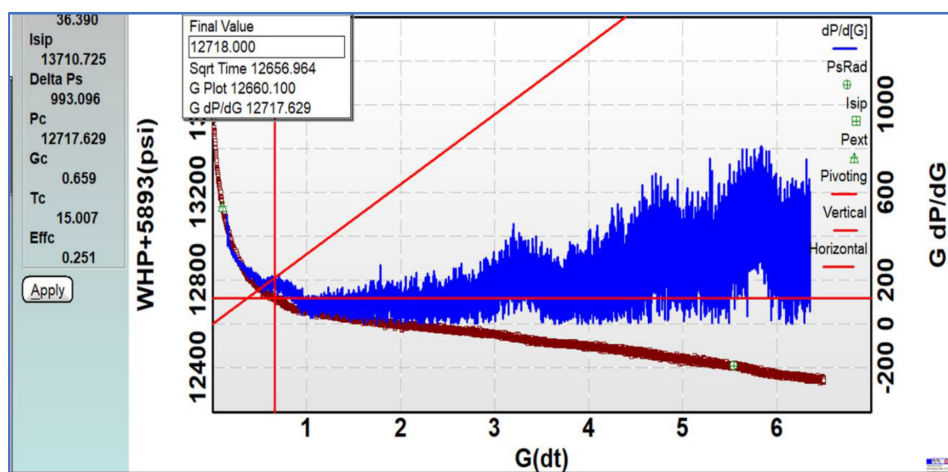
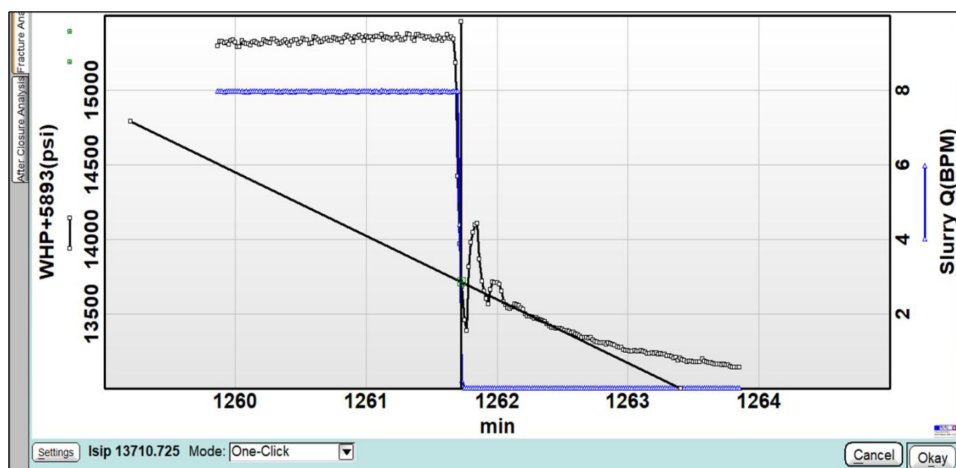


Figure 8—Well A Minifrac Analysis

## MainFrac

In this acid fracturing operations, Foam-Assisted acid systems typically consist of 10–20% nitrogen gas by volume and 80–90% aqueous acid phase, stabilized with surfactants and, in some cases, gelling agents and fluid-loss control additives. These systems are especially effective in stimulating low-permeability carbonate reservoirs and formations, where conventional acid systems may cause excessive fluid leakoff or formation damage. The entrained nitrogen provides an inherent energizing effect, elevating surface treating pressures and enhancing acid penetration into the fracture network. The reduced hydrostatic head of Foam-Assisted acid further minimizes fluid loss, mitigates the risk of mineral sensitivity damage, and facilitates rapid post-treatment cleanup. Compared to low-viscosity fluids, Foam-Assisted acid fracturing reduces total fluid usage while achieving greater fracture length and improved fracture width, resulting in optimum fracture geometry and superior etched-face conductivity.

A diversion methodology was implemented to maximize acid coverage. In the initial stage, an aqueous solution containing a foam-generating surfactant was pumped into the formation and displaced using nitrogen gas. A fraction of the surfactant adsorbed onto fracture faces and the surrounding rock matrix. Subsequent nitrogen injection interacted with the retained surfactant, initiating foam generation through shear and agitation. As formed, its mobility decreased significantly due to its high apparent viscosity and multiphase nature, increasing resistance to flow and gradually raising injection pressure. Once the pressure exceeded the breakdown pressure of an untreated portion of the formation, fluid flow diverted away from the previously stimulated interval, directing acid into a new section of the reservoir. This staged diversion process was repeated to ensure acid distribution across the full treatment interval.

The pad injected into the formation, where its multiphase gas-liquid nature increased flow resistance and bottomhole pressure, further promoting fracture initiation in unstimulated intervals and enabling sequential acid fracturing along the targeted zone. The N<sub>2</sub> fluid is high-viscosity and promotes higher height/length growth with improved fracture width relative to low viscosity completion fluids. These properties of N<sub>2</sub> foam fluids help prevent damage and result in an ideal fracture geometry (Li, N., et al. 2023). The dissolution of carbonate rock is controlled by reactivity, which is greatly reduced after foaming for the same acid strength and temperature. Foamed acid could retard 60–70% acid reactivity. Excessive fluid loss is one of the challenges in acid fracturing. Conventionally, fluid loss is mitigated by using synthetic polymers to viscosify acid that controls leakoff by depositing a low permeability filter cake on the face of formation leading formation damage concerns. Foamed acid does not build filter cake and showed excellent leakoff control (Karadkar, P., et al. 2018).

The modified Single-phase retarded acid systems (mSPRAS) provide higher performance and operational efficiency benefits than systems based on unmodified HCl and emulsified acid. When combined with diverting fluid systems and modeling, they have proven to resolve many challenges with tight carbonate acidfracs. The new single-phase retarded acid has a low viscosity, close to the viscosity of unmodified HCl, which should result in low friction pressure and high carbonate dissolving power (Danican S. et.al, 2024). The success of acid fracturing in tight carbonate formations is heavily dependent on the ability of live acid (unspent acid) to penetrate deeply into the fracture. This deep penetration is necessary to create effective conductivity and stimulate a larger reservoir volume. Through sensitivity analysis, the optimization of various treatment parameters, such as injection rate, acid concentration, and injection strategy, to achieve maximum productivity enhancement is essential- to tailor the acid fracturing design to these specific reservoir conditions (Wang Z. et.al, 2014).

Following comprehensive diagnostic testing and treatment evaluation, the MainFrac design was optimized to maximize fracture conductivity and stimulation effectiveness. Key treatment parameters—including pumping rate, fluid volumes, and acid concentrations—were strategically adjusted to improve conductivity and mitigate the high leakoff behavior observed in the formation (Table 4, Figure 9). A consistently high pumping rate was maintained to sustain net pressure and promote fracture propagation. Advanced fracture modeling and simulation were applied to predict fracture geometry, growth trends,

and fluid distribution, enabling data-driven refinement of the fracture design and fluid system to enhance reservoir contact and production. Post-DFIT analysis, the simulation model was recalibrated using the pressure response to improve predictive accuracy. The final MainFrac design was engineered to achieve the target fracture half-length and width, ensuring effective wormhole development and maximizing overall fracture conductivity (Table 5).

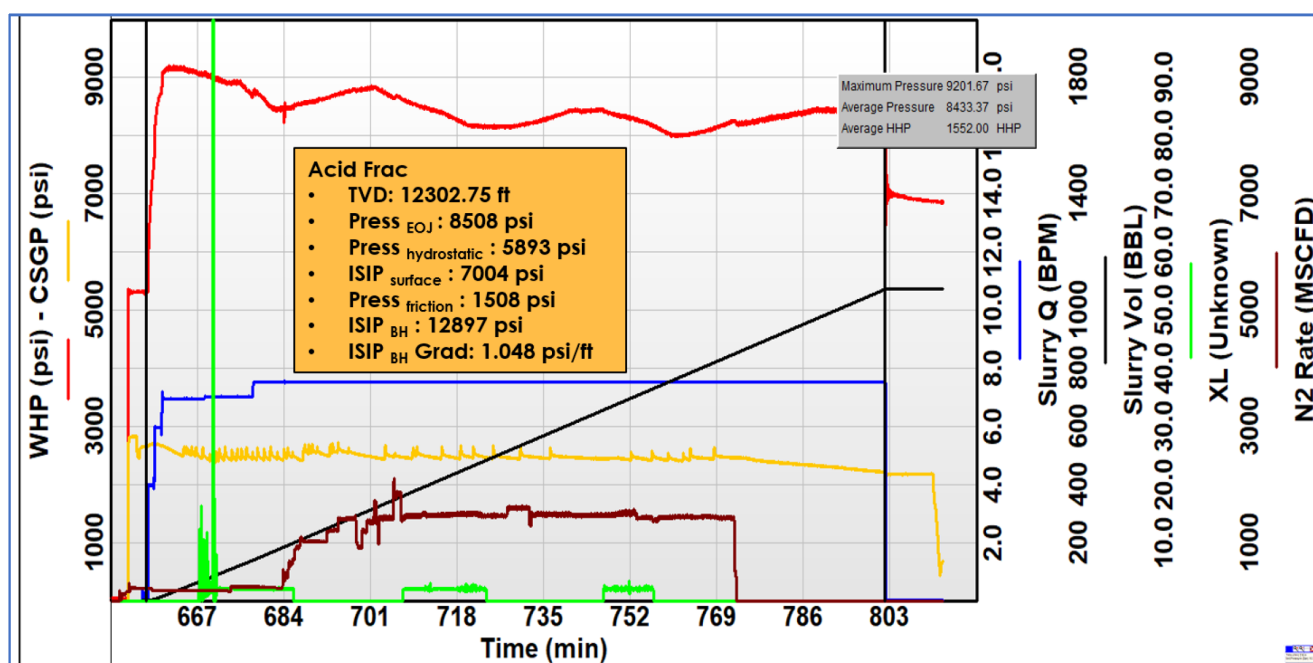


Figure 9—MainFrac Plot

Table 4—MainFrac Schedule

| Step Number | Step Name | Fluid Name | BH Slurry Rate (bpm) | Surf Slurry Rate (bpm) | N2 Rate (scf/min) | B.H Foam Quality Check (%) | Step Slurry Vol. (bbl) |
|-------------|-----------|------------|----------------------|------------------------|-------------------|----------------------------|------------------------|
| 1           | PREFLUSH  | MS         | 8                    | 8.0                    | 0                 | 0%                         | 50                     |
| 2           | PAD-1     | HVFR       | 8                    | 7.2                    | 1396              | 10%                        | 150                    |
| 3           | ACID-1    | mSPARS     | 8                    | 7.2                    | 1396              | 10%                        | 100                    |
| 4           | DIV-1     | VES        | 8                    | 7.2                    | 1396              | 10%                        | 50                     |
| 5           | PAD-2     | HVFR       | 8                    | 7.2                    | 1396              | 10%                        | 150                    |
| 6           | ACID-2    | mSPARS     | 8                    | 7.2                    | 1396              | 10%                        | 100                    |
| 7           | DIV-2     | VES        | 8                    | 7.2                    | 1396              | 10%                        | 50                     |
| 8           | PAD-3     | HVFR       | 8                    | 7.2                    | 1396              | 10%                        | 100                    |
| 9           | ACID-3    | mSPARS     | 8                    | 7.2                    | 1396              | 10%                        | 100                    |
| 10          | POST-FL   | MS         | 8                    | 8.0                    | 0                 | 0%                         | 100                    |
| 11          | FLUSH     | 2% KCL     | 8                    | 8.0                    | 0                 | 0%                         | 120                    |

Table 5—MainFrac Simulation Results

| Hydraulic Fracturing Results w.r.t Ports |                                  |                       |                       |                       |                       |
|------------------------------------------|----------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| PAD Type                                 |                                  | HVFR + N2             | HVFR + N2             | HVFR + N2             | HVFR + N2             |
| Perf #                                   | -                                | Perf #1               | Perf #2               | Perf #3               | Perf #4               |
| Pump Rate                                | Slurry Rate of Injection (bpm)   | 8                     | 8                     | 8                     | 8                     |
| Perf                                     | Measured Depth (ft)              | 13222.00-<br>13227.00 | 13238.03-<br>13249.00 | 13269.00-<br>13275.00 | 13276.00-<br>13292.00 |
|                                          | True Vertical Depth (ft)         | 12275.28-<br>12279.21 | 12287.90-<br>12296.50 | 12312.12-<br>12316.81 | 12317.59-<br>12330.09 |
| Half Length                              | 'Hydraulic' Length (ft)          | 30.1                  | 242.3                 | 30.5                  | 510.8                 |
|                                          | Acid Etched length (ft)          | 30.1                  | 242.3                 | 30.5                  | 510.8                 |
|                                          | Average Acid Etched Width.(in)   | 0.43                  | 0.27                  | 0.44                  | 0.39                  |
|                                          | Average Conductivity (md-ft)     | 24962.5               | 24530.0               | 19022.1               | 5586.7                |
| HEIGHT:                                  | Max Fracture Height (ft)         | 11.9                  | 25.2                  | 47.4                  | 103.8                 |
| WIDTH:                                   | Avg width at end of pumping (in) | 0.42                  | 0.27                  | 0.44                  | 0.38                  |

## Production Analysis & Conclusions

- Campaign Overview: A total of four wells were stimulated during the campaign.
- Performance Improvement: Comparative analysis of offset wells showed that Foam-Assisted fracturing fluids—formulated with N<sub>2</sub> foams—delivered significantly higher productivity than conventional non-foam acid systems.
- Foam-Assisted Benefits:
  - Reduced fluid leakoff and enhanced acid penetration.
  - Lowered effective fluid density and minimized phase trapping.
  - Improved cleanup efficiency, particularly in tight, low-permeability zones.
- Fracture Geometry: High-resolution logs, BHP data, and sonic anisotropy confirmed fracture confinement within the target zone.
  - Average fracture half-length: ~300 ft.
  - Estimated fracture height: 45–100 ft.
- Permeability Enhancement: Pressure-transient analysis indicated a significant increase in permeability due to:
  - Enlargement of natural fracture apertures.
  - Etched fracture faces from acid-rock reactions.
- Diagnostic Parameters:
  - Instantaneous Shut-In Pressure (ISIP): ~13,700 psi.

- Closure pressure: ~12,800 psi.
- G-function and  $\sqrt{t}$  pressure decline analysis indicated a complex but well-connected fracture network with minimal near-wellbore tortuosity.
- Flowback Efficiency: Foam-Assisted system enabled rapid fluid recovery with >80% flowback efficiency.
- Production Results:
  - Well A achieved a four-fold increase in daily production rate.
  - Initial post-frac rates were sustained for over 60 days before gradual decline, confirming effectiveness and durability of the treatment.

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